Business Requirements Document (BRD)

Digital Loan Application Process Improvement

Prepared by: Jeevika CP, Business Analyst (portfolio project) | **Version:** 1.0 | **Date:** 22 October 2026

> **Data note:** This is an independent portfolio project. The analysis uses a simulated dataset of 12,000 loan applications (Jan-Dec 2025) built for this project; it is not real bank data. All figures are illustrative, and projected impacts are assumption-based estimates, not guarantees. Currency is in ₹.

1. Executive Summary

A retail bank's digital loan application process loses 37.1% of applicants before a decision, and approved applicants wait an average of 10.8 days. Analysis of 12,000 applications shows the biggest problem is the **document upload stage**, which accounts for 41.4% of all abandonments and about ₹2,812 million in loan value. Applications with incomplete documents on the first attempt are abandoned 50.3% of the time, against 30.6% for complete ones, and mobile app users are hit hardest.

This document defines the requirements for a redesigned process built around guided document upload with real-time validation, automated verification and a transparent status tracker. If the project recovers 30% of document-stage abandonments, the model projects about 423 additional approved loans a year (approval rate 54.0% to 57.5%), worth roughly ₹577 million in extra approved loan value.

2. Business Problem and Opportunity

Problem statement: Customers who start a digital loan application frequently abandon it because uploading documents is error-prone and verification is slow and opaque. This causes lost revenue, wasted processing effort and a poor customer experience.

Opportunity: Fixing the document and verification stages addresses 62% of all abandonments (41.4% at document upload plus 20.8% at verification), using improvements that are well understood in digital banking.

3. Business Objectives and Success Metrics

ID	Objective	Baseline	Target
BO-1	Reduce overall abandonment rate	37.1%	30% or lower within 12 months of launch
BO-2	Reduce abandonment at document upload	1,842 applications (17.9% of those reaching the stage)	At least 30% reduction
BO-3	Reduce average days to decision	10.8 days	8 days or fewer
BO-4	Increase documents complete on first attempt	67.0%	85% or higher
BO-5	Increase approval rate	54.0%	57% or higher
BO-6	Reduce mobile app abandonment	42.1%	Match or beat branch (31.4%)

4. Scope

In scope: Web and mobile application journey for Personal, Home, Auto and Education loans; document upload and validation; identity, employment and income verification; application status tracking and notifications; management dashboard of KPIs.

Out of scope: Changes to credit scoring models and lending policy; loan servicing and collections; branch-staff systems beyond the verification work queue; marketing and acquisition.

5. Stakeholders

Stakeholder	Role in project	Key interest
Head of Retail Lending	Sponsor, approves scope and budget	Higher disbursals, lower cost per loan
Loan Operations Manager	Process owner	Fewer manual checks and rework
Credit Risk Manager	Reviewer	No weakening of risk controls
Compliance and KYC Officer	Reviewer	Regulatory and audit compliance
IT / Digital Product Team	Delivery	Feasible, secure integration
Customer Service Lead	Consulted	Fewer status and document queries
Loan Applicants	End users	Fast, clear, easy application

6. Current State (AS-IS) Analysis

Analysis was done with SQL and Python on the application dataset. See the AS-IS process map and KPI dashboard in the project folder.

#	Finding	Evidence
1	Document upload is the largest drop-off point	41.4% of abandonments (1,842 of 4,454)
2	Incomplete documents sharply raise abandonment and delay	50.3% vs 30.6% abandonment; 14.5 vs 9.5 days to decision
3	Only 67.0% of applications have complete documents first time	Branch 85.6%, Online 60.7%, Mobile App 51.1%
4	Mobile app and online channels underperform branches	Abandonment 42.1% mobile, 39.0% online, 31.4% branch
5	Manual verification slows non-salaried applicants	12.6 days (self-employed) vs 9.6 days (salaried)
6	Longer verification waits increase drop-off	Abandoned at verification: 9.0% (1-4 days), 11.4% (5-7), 15.2% (8+)
7	Document problems cause rejections too	23.7% of rejections are for incomplete or invalid documents

Root causes (summary): no real-time document validation; unclear checklists and poor image capture on mobile; staff-dependent manual verification; no status visibility, so customers cannot see progress and drop off.

7. Proposed Solution (TO-BE)

1. **Instant eligibility check and shorter form** with pre-filled details.
2. **Guided document upload** with a checklist, image-quality checks and instant on-screen fix prompts.
3. **Automated e-KYC and income verification**, with staff reviewing exceptions only.
4. **Status tracker with SMS and app notifications** at every stage.
5. **Save and resume** so applicants can finish later on any device.
6. **KPI dashboard** for operations and management.

See the TO-BE process map and wireframes in the project folder.

8. Functional Requirements

Priority uses MoSCoW (M = Must, S = Should, C = Could).

ID	Requirement	Priority	Objective
FR-01	System shall show an indicative eligibility result from loan type, amount, income and employment before the full form	S	BO-1
FR-02	System shall pre-fill known customer details for existing customers	S	BO-1
FR-03	System shall display a document checklist specific to loan type and employment type	M	BO-4
FR-04	System shall validate each uploaded document in real time for format, size, legibility and expiry, and show the result immediately	M	BO-2, BO-4
FR-05	System shall tell the applicant exactly what to fix and allow re-upload in the same session	M	BO-2
FR-06	System shall support mobile camera capture with edge detection and light and blur guidance	S	BO-6
FR-07	System shall verify identity through e-KYC integration without manual steps where data matches	M	BO-3
FR-08	System shall verify salaried income automatically using uploaded statements and employer data, and route only mismatches to staff	M	BO-3
FR-09	System shall provide a status tracker showing the current stage and expected decision date	M	BO-1
FR-10	System shall send SMS or app notifications on each stage change and on any required action	M	BO-1
FR-11	System shall let applicants save progress and resume on any device	S	BO-1, BO-6
FR-12	System shall send an automatic reminder if an application is inactive for 48 hours	S	BO-1
FR-13	System shall provide a staff work queue for exception cases with time limits and escalation	M	BO-3
FR-14	System shall provide a dashboard of funnel, abandonment, processing time and rejection reasons by channel, loan type and month	S	All
FR-15	System shall offer a chat or callback option to a loan advisor from any step	C	BO-1

9. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All documents encrypted in transit and at rest; role-based access; full audit trail
NFR-02	Compliance	Meets KYC, data privacy and consent requirements; explicit applicant consent recorded
NFR-03	Performance	Document validation result in 5 seconds or less; pages load in 3 seconds or less on 4G
NFR-04	Availability	99.5% uptime during business hours
NFR-05	Usability	Works on current mobile browsers and the app; accessible and available in English and Tamil at launch
NFR-06	Scalability	Handles at least 3x current monthly application volume
NFR-07	Data retention	Application data retained and purged per bank policy

10. User Stories and Acceptance Criteria

US-01 Eligibility check. As an applicant, I want to see if I am likely eligible before filling the full form, so that I do not waste time. *Acceptance:* Given I enter loan type, amount, income and employment, when I tap Check, then I see an indicative eligibility result within 3 seconds, clearly marked as indicative.

US-02 Document checklist. As an applicant, I want to see exactly which documents I need, so that I upload the right ones first time. *Acceptance:* Given I select a loan and employment type, when I reach the upload step, then I see a checklist of only the required documents with accepted formats.

US-03 Real-time validation. As an applicant, I want to know immediately if a document is unusable, so that I can fix it without waiting days. *Acceptance:* Given I upload a blurry, expired or wrong-format document, when validation runs, then I see the specific problem and a Retake button within 5 seconds.

US-04 Mobile capture. As a mobile user, I want guidance when photographing documents, so that the image is accepted. *Acceptance:* Given I use the camera, when I frame a document, then the app shows edge detection and warns about low light or blur before capture.

US-05 Automated verification. As a loan officer, I want verified applications to move forward automatically, so that I only handle exceptions. *Acceptance:* Given all data matches, when verification runs, then the application proceeds to credit assessment with no manual task; mismatches appear in my work queue with reasons.

US-06 Status tracking. As an applicant, I want to see where my application is and when to expect a decision, so that I do not need to call the bank. *Acceptance:* Given I log in, when I open my application, then I see the current stage, completed steps and an expected decision date.

US-07 Notifications. As an applicant, I want alerts when something changes or I must act, so that nothing stalls. *Acceptance:* Given my status changes or an action is needed, when the event occurs, then I receive an SMS or app notification within 5 minutes.

US-08 Save and resume. As an applicant, I want to save my application and continue later, so that I can finish when I have my documents. *Acceptance:* Given I exit midway, when I log in again on any device, then my entered data and uploaded documents are restored.

US-09 Inactivity reminder. As a sales manager, I want inactive applicants reminded, so that we recover lost applications. *Acceptance:* Given an application is inactive for 48 hours, when the threshold passes, then one reminder is sent with a link to resume.

US-10 Management dashboard. As the Head of Retail Lending, I want a live dashboard of funnel and processing KPIs, so that I can spot problems early. *Acceptance:* Given I open the dashboard, when I filter by channel, loan type or month, then I see approval rate, abandonment rate, drop-off by stage and average days to decision.

11. Business Case and Projected Impact

The model assumes a share of document-stage abandonments are recovered and then convert like other applicants who reached verification (76.5% approval). Average approved loan size in the data is ₹1.36 million.

Scenario	Extra approved loans	Approval rate after	Extra approved value
Recover 20% of document-stage abandonments	282	56.4%	₹385 million
Recover 30% (base case)	423	57.5%	₹577 million
Recover 40%	564	58.7%	₹769 million

The base case also lowers overall abandonment from 37.1% to about 32.5%; reaching the 30% target depends on the verification improvements as well. Benefits not quantified here: lower manual processing effort, fewer status calls and higher customer satisfaction. Costs (development, integration and licences) must be estimated with IT before approval.

12. Assumptions, Constraints and Risks

Assumptions: the bank can integrate with an e-KYC provider; recovered applicants behave like current applicants; lending policy remains unchanged.

Constraints: regulatory KYC and data privacy rules; legacy loan system integration; limited IT capacity.

Risk	Impact	Mitigation
Automated validation wrongly rejects genuine documents	Frustrated applicants	Allow manual override and tune thresholds in a pilot
Fraud rises with faster approvals	Credit loss	Keep risk checks unchanged; monitor fraud rate as a KPI
Low adoption of new features	Benefits not realised	Pilot with one channel; user testing; staff training
Integration delays with the legacy system	Schedule slip	Phase delivery; use APIs and a staging environment early

13. Implementation Roadmap

Phase	Scope	Indicative timeline
1	Document checklist, real-time validation, mobile capture, reminders (FR-03 to FR-06, FR-12)	Months 1-3
2	Automated e-KYC and income verification, exception work queue (FR-07, FR-08, FR-13)	Months 3-6
3	Status tracker, notifications, save and resume, eligibility check, dashboard (FR-01, FR-02, FR-09 to FR-11, FR-14, FR-15)	Months 6-9

Success is reviewed monthly against the KPIs in section 3. Pilot on the mobile app first, since it has the highest abandonment.

14. Approval

Name	Role	Signature	Date
	Head of Retail Lending		
	Loan Operations Manager		
	Credit Risk Manager		
	Compliance Officer		

Appendix: Data and Method

Dataset: 12,000 simulated applications with channel, loan type, amount, employment type, document completeness, stage reached, outcome, rejection reason, verification days and days to decision. Analysis: 10 SQL queries (SQLite) run through Python and visualised in a KPI dashboard. All code, data and charts are in the project folder.